DATA ANALYTICS PORTFOLIO REPORT

Telecom Customer Churn Analysis

SQL Server + Power BI | Customer Churn · Retention Risk · Revenue Loss

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1. Executive Summary

- The project analyzes telecom customer churn and the revenue associated with churned versus retained customers.
- The SQL layer covers data-quality checks, NULL handling, text trimming, duplicate detection, categorical validation, KPI calculations, customer segmentation, tenure analysis, and revenue lost to churn.
- The Power BI dashboard summarizes the analysis through KPI cards, churn distribution, contract, tenure, internet-service, payment-method, and monthly-charge visuals plus slicers.
- The dashboard reports 7,043 customers, 1,869 churned customers, a 26.54% churn rate, 16.06M total revenue, 64.76 average monthly charge, and 2.86M lost to churn.

2. Business Problem

A telecom provider needs to understand why customers leave, which customer/service segments carry the highest churn exposure, and where churn is creating revenue loss so retention activity can be prioritized.

3. Project Objectives

- Validate the customer-level data before analysis.
- Clean selected missing and whitespace-affected values.
- Calculate customer, churn, revenue, and average-charge KPIs.
- Compare churn across demographics, contracts, services, payment methods, and tenure.
- Quantify revenue associated with churned customers.
- Present the findings in an interactive Power BI dashboard.

4. Dataset Overview

The telecom churn dataset contains **7,043 customer records** (verified directly against the supplied CSV) and includes demographic, service usage, contract, payment, billing, and churn information. This figure matches the dashboard's Total Customers KPI exactly.

- Customer ID and demographic attributes
- Partner and dependent status
- Tenure
- Phone and internet services
- Additional service features
- Contract and payment method
- Monthly charges and total charges

- Churn status

5. SQL Analysis

EDA & Data Cleaning

- Inspected the table with SELECT *.
- Checked row volume and NULLs in customerID, TotalCharges, and MultipleLines.
- Replaced NULL TotalCharges values with 0.
- Trimmed whitespace from Partner, Dependents, PhoneService, and Churn.
- Checked duplicate customer IDs using GROUP BY and HAVING.
- Validated distinct values for gender, Contract, InternetService, and Churn.
- Summarized churned versus retained customers.

Analytical SQL

- COUNT, SUM, and AVG aggregations
- CASE-based churn counting and percentage calculation
- GROUP BY and HAVING segmentation
- Contract, service, payment, and tenure analysis
- Revenue split between churned and retained customers

Advanced SQL

The supplied SQL does not contain JOINs, CTEs, window functions, ranking functions, running totals, or cumulative percentage calculations. Documenting these as project techniques would therefore overstate the actual SQL work.

6. Important SQL Techniques

Technique	Evidence in supplied SQL
Data-quality validation	COUNT + conditional NULL checks
Cleaning	UPDATE + LTRIM/RTRIM
Duplicate detection	GROUP BY + HAVING
Categorical validation	SELECT DISTINCT
KPI calculation	COUNT, SUM, AVG, CASE, CAST
Segmentation	GROUP BY across customer/service fields
Revenue analysis	Conditional SUM by Churn

7. Power BI Development

- The supplied PBIX contains one report page.
- The model metadata shows one table and no auto-created relationships.

- The report contains KPI measures named Total_Customers, Total_Churned, Churn_Rate %, Total_Revenue, Avg_Monthly_Charge, and Revenue_Lost.
- The dashboard includes KPI cards, donut, column, scatter, line, clustered bar, ribbon, and slicer visuals.
- Slicers are provided for Contract, Internet Service, and Gender.
- Additional dashboard fields include Contract Short and TenureGroup.

The exact DAX expressions behind the measures are not reproduced because the accessible PBIX metadata exposes the measure references used by visuals but not reliable underlying formulas. This avoids inventing DAX that was not directly available in the supplied materials.

8. Dashboard KPIs

KPI	Dashboard value
Total Customers	7,043
Churned Customers	1,869
Churn Rate	26.54%
Total Revenue	16.06M
Average Monthly Charge	64.76
Revenue Lost to Churn	2.86M

9. Key Business Insights

- Monthly contracts are the most exposed segment in the dashboard, explicitly highlighted at 44% churn.
- Electronic Check is explicitly highlighted as the highest churn-risk payment method.
- Fiber optic has the highest churn count among the displayed internet-service categories.
- The dashboard highlights a visible relationship between higher monthly charges and churn.
- The tenure visualization shows heavier churn among newer customers, with much stronger retained-customer volume at longer tenure.

10. Business Recommendations

- Prioritize retention offers and contract-conversion campaigns for monthly-contract customers.
- Investigate the customer experience and billing journey associated with Electronic Check users.
- Review the Fiber Optic customer journey, onboarding, service quality, and support experience for churn triggers.
- Use charge-sensitive retention segmentation to identify higher-value customers showing churn risk.
- Strengthen first-year onboarding and early-tenure engagement, where the dashboard shows the greatest churn concentration.

11. Results

The analysis establishes a clear retention-risk profile across contract type, payment method, internet service, monthly charges, and tenure. The dashboard converts these patterns into an executive-friendly view that can support targeted retention planning.

12. Conclusion

This project demonstrates practical end-to-end analytics skills: preparing and validating customer data in SQL Server, calculating business KPIs, segmenting churn and revenue, and presenting the results in Power BI. It is intentionally documented without claiming advanced SQL or DAX techniques that are not supported by the supplied materials.

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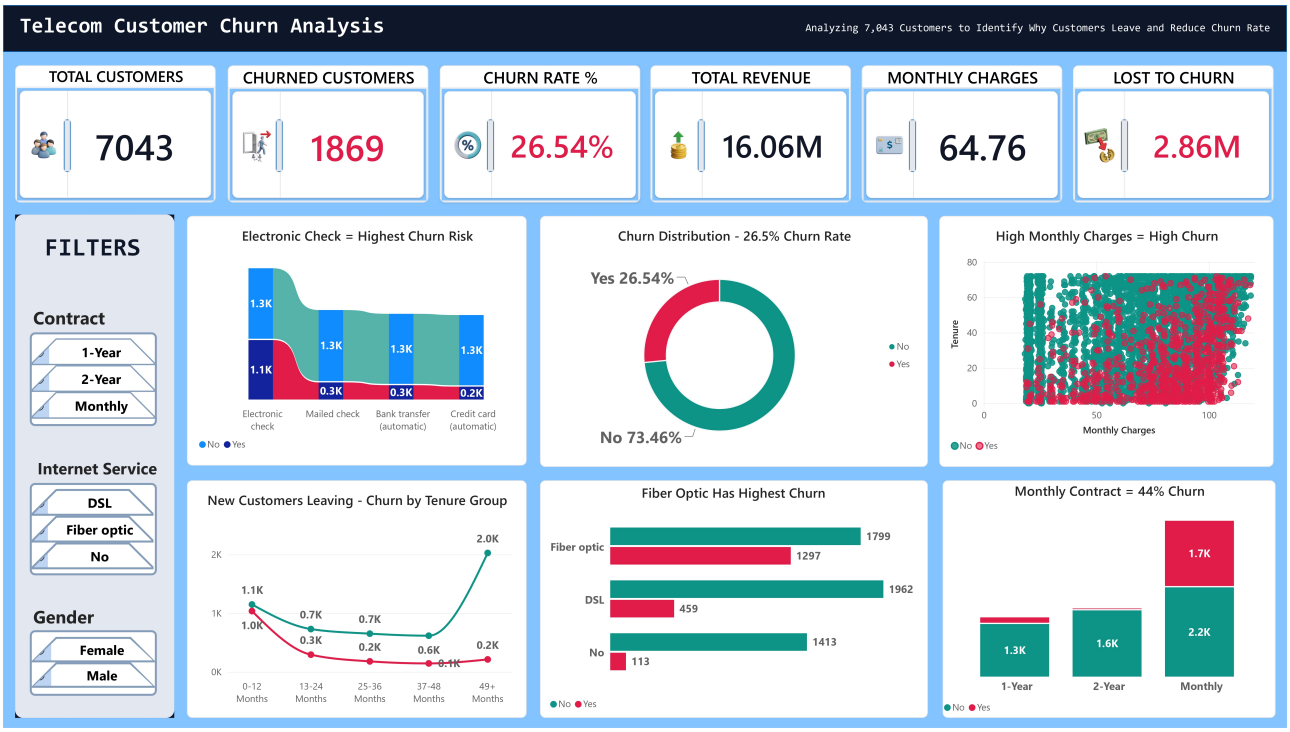
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This report accompanies the full SQL scripts, Power BI (.pbix) file, source dataset, and README documentation available in the project repository.

13. Dashboard Screenshot

The following screenshot is the supplied dashboard image and is included as the visual source of truth for the dashboard section of this report.



Source: supplied Power BI dashboard screenshot.